

Jialong Fan

Third-Year Undergraduate, Physics + Computer Science, The Hong Kong University of Science and Technology |
jfanaw@connect.ust.hk

PhD Research Direction

Prospective PhD applicant broadly interested in quantum computing and its theoretical foundations, including quantum algorithms, quantum cryptography, fault-tolerant architectures, and mathematical structures connecting coding theory, information theory, and computation. Current preparation includes stabilizer/CSS codes, qLDPC/qLTC codes, high-dimensional expanders, decoder baselines, and simulation practice.

Education

The Hong Kong University of Science and Technology, Undergraduate Student in Physics + Computer Science 2023 - Present

- Third-year undergraduate with cross-disciplinary training in physics and computer science.
- Academic interests lie at the interface of quantum theory, algorithms, secure information processing, mathematical structures, and computational methods.

Research Preparation and Academic Experience

Study in Quantum Error Correction and FTQC, Shilin Huang Group Junior Year - Present

- Studying fault-tolerant quantum computation and quantum error correction topics with exposure to qLDPC codes, toric codes, BP-OSD decoders, and related variants.
- Developing a theoretical foundation for code constructions, decoding algorithms, and their roles in scalable fault-tolerant architectures.

UROP Learning Experience in NV-Center Quantum Computing, Yang Sen Group, HKUST Sophomore Fall

- Gained introductory exposure to NV-center-based quantum computing through an HKUST UROP project, building context for physical platforms and quantum information applications.

Independent Study and Research Direction

- Independently studying high-dimensional expanders and their connections to quantum locally testable codes, with attention to locality, testability, distance, rate, and decodability.
- Particularly drawn to broad, mathematically structured problems in quantum computing, including secure protocols, algorithms, fault tolerance, and information-processing tasks.
- Comfortable learning across physics, mathematics, and computer science, and motivated to connect formal theory with simulation and algorithmic understanding.

Technical Skills

Programming	Python; basic C++; R.
Quantum Simulation	Experience using the Stim Python library for quantum-circuit and quantum-error-correction simulations.
Relevant Topics	FTQC; qLDPC codes; toric codes; BP-OSD decoders and variants; high-dimensional expanders; qLTC.

Language

TOEFL iBT: 113

Personal Statement

Jialong Fan | jfanaw@connect.ust.hk

I am a third-year undergraduate student at the Hong Kong University of Science and Technology, majoring in Physics and Computer Science. My current academic interests lie in broad quantum computing and its theoretical foundations, with a particular interest in quantum cryptography, secure quantum protocols, and related problems where quantum information, algorithms, and coding theory interact.

My previous research preparation has been centered on quantum error correction and fault-tolerant quantum computation. In Shilin Huang's group, I have studied stabilizer/CSS codes, toric codes, qLDPC codes, BP-OSD-style decoders, and homological viewpoints on quantum codes. I have also used Python and Stim to simulate a flagged 7-qubit color code under circuit-level noise, and reproduced a BP-SF decoder baseline as an implementation exercise. These experiences are not cryptography projects by themselves, and I do not want to overstate the connection. Rather, I see them as preparation in the language of quantum information: stabilizer formalisms, noisy quantum processes, coding-theoretic thinking, verification-style questions, and rigorous reasoning about how information is represented, protected, or inferred in quantum systems. I hope to build on this foundation when moving toward quantum cryptography and secure quantum protocols.

I am especially interested in mathematically structured and proof-oriented research. In addition to group reading, I have been independently studying high-dimensional expanders and their connections to quantum locally testable codes. This has helped me build familiarity with the language of chain complexes, homology/cohomology, expansion, locality, distance, and testability. More importantly, it gave me a route into theoretical cryptography through the classical side of local testing: expansion is a central tool in PCP theory, for example in Dinur's gap-amplification proof of the PCP theorem, and PCPs later became a foundation for succinct cryptographic proof systems such as Kilian-style arguments and Micali's CS proofs. Following this path from coding and expansion to PCPs and then to cryptographic proof systems is one reason my interest has moved toward theoretical cryptography. I am still at an early stage, but this independent study reflects the way I hope to grow as a researcher: by reading beyond coursework, organizing unfamiliar mathematical tools, and turning abstract ideas into concrete examples or technical notes.

My transcript should also be read in the context of a heavy course load. Since my sophomore year, I have often taken substantially more than the regular 18-credit load; for example, I took 25 credits in 2025-26 Fall. This overload affected my cumulative GPA, especially because some programming and systems-oriented courses were weaker than my theoretical courses. At the same time, my transcript shows stronger performance in more theoretical subjects, including A+ grades in COMP2711 Discrete Mathematical Tools for Computer Science and COMP3711 Design and Analysis of Algorithms. I do not view the course load as an excuse, but I believe it gives useful context for understanding my preparation and the distribution of my strengths.

I hope to develop my background toward quantum cryptography and theoretical quantum computing. My physics training gives me the language for quantum systems, my computer science training gives me algorithmic and complexity-oriented habits, and my recent QEC and HDX/qLTC preparation gives me a concrete starting point for rigorous work involving quantum information, coding, security, and mathematical proof. If appropriate, I would also be very interested in starting with a sustained reading/research project or RA-style engagement during my final undergraduate year, as a way to build stronger preparation and establish research fit before doctoral study. I believe this combination is a good fit for research groups working on quantum cryptography and the theoretical foundations of quantum computation.

Name:

Student ID:

Year of Study:

Print Date:

FAN, Jialong 范嘉龙

21041112

3rd Year

8 June 2026

02:49:44

Reg Status:

Active in Program

Unofficial Transcript of Academic Record (as of 08 June 2026)

Advisor(s):

HAN, Yilong

PAPADOPOULOS, Dimitris

Academic Program

Admit Date:

1 September 2023

Program:

Bachelor Degree in School of Science

Program Change:

2024-25 Fall

Program:

Bachelor of Science

Major:

Physics

Program Change:

2024-25 Fall

Program:

Bachelor of Science

Major:

Physics (with Extended Major in Artificial Intelligence)

Effective:

2024-25 Spring

Add Major:

Computer Science

On Exchange:

2025-26 Spring

Technische Universität München

Academic Records

2023-24 Fall (Full-Time)

Course Code	Course Title	Credit Attempted	Credit Earned	Grade
HMAW1905A	Behavioral Foundations of University Education: Habits, Mindsets, and Wellness	3.0	-	PP
LANG1402	English Language for University Studies	3.0	3.0	B+
LEGL1000	Legal Education (PRC Constitution, Basic Law, National Security Law and Legal System of HKSAR)	0.0	-	P
LIFS1902	General Biology II	3.0	3.0	A
MATH1023	Honors Calculus I	3.0	3.0	A-
PHYS1113	Laboratory for General Physics I	1.0	1.0	B+

Course Code	Course Title	Credit Attempted	Credit Earned	Grade
PHYS1312	Honors General Physics I	3.0	3.0	A-
PHYS2022	Modern Physics	3.0	3.0	A+

TGA: 3.769

CGA: 3.769

Cumulative Credits Earned: 16.0

Awards:

Dean's List

2023-24 Spring (Full-Time)

Course Code	Course Title	Credit Attempted	Credit Earned	Grade
CHEM1020	General Chemistry I	3.0	3.0	A
HMAW1905A	Behavioral Foundations of University Education: Habits, Mindsets, and Wellness	3.0	3.0	P
LANG1403S	Academic English for University Studies	3.0	3.0	A-
MATH1024	Honors Calculus II	3.0	3.0	A
PHYS1115	Laboratory for General Physics II	1.0	1.0	A
PHYS1314	Honors General Physics II	3.0	3.0	B
SCIE1500	Guided Study on Research I	1.0	1.0	A

TGA: 3.721

CGA: 3.747

Cumulative Credits Earned: 33.0

Awards:

Dean's List

2023-24 Summer (Full-Time)

Course Code	Course Title	Credit Attempted	Credit Earned	Grade
HUMA2101	Enjoyment of Western Opera	3.0	3.0	A-
SOSC1969	Discovering Mind and Behavior	3.0	3.0	A-

Name: FAN, Jialong 范嘉龙
Student ID: 21041112
Year of Study: 3rd Year
Print Date: 8 June 2026
02:49:44

Reg Status: Active in Program

Unofficial Transcript of Academic Record (as of 08 June 2026)

Course Code	Course Title	Credit Attempted	Credit Earned	Grade
-------------	--------------	------------------	---------------	-------

TGA: 3.700

CGA: 3.739

Cumulative Credits Earned: 39.0

2024-25 Fall (Full-Time)

Course Code	Course Title	Credit Attempted	Credit Earned	Grade
COMP1021	Introduction to Computer Science	3.0	3.0	A-
COMP2711	Discrete Mathematical Tools for Computer Science	4.0	4.0	A+
ELEC1010	Electronic and Information Technology	3.0	3.0	A
MATH2121	Linear Algebra	4.0	4.0	A
PHYS2023	Modern Physics Laboratory	1.0	1.0	A+
PHYS2080	Physics Seminar and Tutorial I	1.0	1.0	P
PHYS3038	Optics	3.0	3.0	A
PHYS4050	Thermodynamics and Statistical Physics	3.0	3.0	A+
UROP1100N	Undergraduate Research Opportunities Series 1	1.0	1.0	P

TGA: 4.071

CGA: 3.861

Cumulative Credits Earned: 62.0

Awards: Dean's List
Physics Entry Scholarship
University's Scholarship Scheme for Continuing Undergraduate Students

2024-25 Winter (Full-Time)

Awards: Physics Entry Scholarship
University's Scholarship Scheme for Continuing Undergraduate Students

2024-25 Spring (Full-Time)

Course Code	Course Title	Credit Attempted	Credit Earned	Grade
COMP2011	Programming with C++	4.0	4.0	B-
MATH2043	Honors Mathematical Analysis	4.0	4.0	B
PHYS2124	Mathematical Methods in Physics I	3.0	3.0	A
PHYS3032	Classical Mechanics	3.0	3.0	A
PHYS3034	Electricity and Magnetism II	3.0	3.0	A-
PHYS3037	Honors Quantum Mechanics I	4.0	4.0	B+
PHYS3152	Methods of Experimental Physics I	3.0	3.0	B+

TGA: 3.375

CGA: 3.717

Cumulative Credits Earned: 86.0

Awards: Physics Entry Scholarship
University's Scholarship Scheme for Continuing Undergraduate Students

2024-25 Summer (Full-Time)

Course Code	Course Title	Credit Attempted	Credit Earned	Grade
MATH2411	Applied Statistics	4.0	4.0	A

TGA: 4.000

CGA: 3.731

Cumulative Credits Earned: 90.0

Awards: Physics Entry Scholarship
University's Scholarship Scheme for Continuing Undergraduate Students

2025-26 Fall (Full-Time)

Course Code	Course Title	Credit Attempted	Credit Earned	Grade
COMP2012	Object-Oriented Programming and Data Structures	4.0	4.0	C

Name: FAN, Jialong 范嘉龙
Student ID: 21041112
Year of Study: 3rd Year
Print Date: 8 June 2026
02:49:44

Reg Status: Active in Program

Unofficial Transcript of Academic Record (as of 08 June 2026)

Course Code	Course Title	Credit Attempted	Credit Earned	Grade
COMP2611	Computer Organization	4.0	4.0	B-
COMP3711	Design and Analysis of Algorithms	3.0	3.0	A+
COMP4900	Academic and Professional Development	0.0	-	PP
LANG2010	English for Science I	3.0	3.0	C
MATH2023	Multivariable Calculus	4.0	4.0	B+
PHYS3053	Honors Electricity and Magnetism I	4.0	4.0	B+
PHYS4051	Quantum Mechanics II	3.0	3.0	A-

TGA: 3.008

CGA: 3.566

Cumulative Credits Earned: 115.0

Awards: Paul Ching Wu Chu Scholarship for Physics Students

2025-26 Winter (Full-Time)

Awards: Paul Ching Wu Chu Scholarship for Physics Students

2025-26 Spring (Full-Time)

Awards: Paul Ching Wu Chu Scholarship for Physics Students

- End of Transcript -